

VirtuaLab Digital

Complete Ecosystem System Architecture

Systematic Blueprint: No Guessing, Just Systems

Baserow Data Layer | n8n Automation Layer | GoHighLevel CRM Layer | AI Agent Orchestration Layer

Document Type	System Architecture Blueprint
Framework	AISEO Framework v3.0 (75-Tab Strategic Blueprint)
Prepared For	VirtuaLab Digital Operations
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Principle	Every action is data-driven, tracked, repeatable, and measurable.

1. System Philosophy: Why Systematic, Not Guessing

The fundamental problem with most digital marketing agencies is that their operations rely on tribal knowledge, individual heroics, and ad-hoc decision making. A senior SEO specialist "knows" what keywords to target based on experience. A content writer "feels" what topics will resonate. An account manager "remembers" to check client rankings every few weeks. This approach is fragile, unscalable, and fundamentally unmeasurable. When the specialist leaves, the knowledge leaves. When the workload doubles, quality halves. When a new client onboards, the process restarts from zero every time.

VirtualLab Digital operates on a different principle: every strategic decision, every content action, every client interaction, and every measurement must flow through a systematic, data-driven infrastructure. This ecosystem blueprint defines the exact data structures (Baserow), automation workflows (n8n), client management pipelines (GoHighLevel), and AI agent orchestration protocols that transform the 75-tab AISEO Framework from a static document into a living, breathing operational machine. The goal is simple: any trained operator should be able to execute the strategy by following the system, regardless of their individual expertise level. The system captures intelligence, automates repetition, enforces consistency, and measures everything.

1.1 The Four-Layer Architecture

The VirtualLab Digital ecosystem is organized into four interdependent layers, each with a distinct technological foundation and operational responsibility. The Data Layer (Baserow) serves as the single source of truth for all operational data including keywords, content, competitors, clients, and performance metrics. The Automation Layer (n8n) connects all systems, automates repetitive workflows, and enforces process consistency. The CRM and Client Delivery Layer (GoHighLevel) manages the client lifecycle from lead capture through onboarding, delivery, and retention. The AI Agent Orchestration Layer coordinates the six specialized AI agents defined in the AISEO Framework, ensuring that each agent receives the correct inputs, produces standardized outputs, and feeds results back into the data layer for continuous learning.

Core Principle: Data enters the system once, flows through automated workflows, triggers agent actions, surfaces in client reports, and feeds back into strategic decisions. No data lives in silos. No action is taken without a data trigger. No decision is made without a measurement checkpoint.

1.2 Data Flow Architecture

Information flows through the ecosystem in a unidirectional pipeline with feedback loops. Raw data enters through web search APIs, SERP monitoring tools, Google Search Console, and manual research inputs. This data is normalized and stored in Baserow tables. n8n workflows poll Baserow for new records, process them through AI agents (via LLM API calls), and write outputs back to Baserow. Client-facing actions (content publishing, report generation, email notifications) are triggered by n8n and executed through GoHighLevel or direct API calls. Performance data from published content flows back through web analytics APIs into Baserow, where it is compared against baselines and triggers optimization workflows. This closed-loop system ensures that every action is informed by data and every outcome is measured against objectives.

Stage	Trigger Source	Destination	Action
1. Data Ingestion	External APIs, Manual Input	Baserow Tables	Keywords, competitors, SERP data, client info stored as structured records.
2. Processing	Baserow (trigger)	n8n Workflows	New records trigger workflows: clustering, content briefs, competitor alerts.
3. Agent Orchestration	n8n (dispatch)	AI Agent APIs	6 specialized agents process tasks: research, positioning, analysis, strategy.

Stage	Trigger Source	Destination	Action
4. Output Generation	AI Agents (return)	Baserow + n8n	Agent outputs stored: content briefs, keyword maps, optimization recommendations.
5. Client Delivery	Baserow (status change)	GHL + n8n	Approved content published, reports generated, clients notified.
6. Performance Feedback	Analytics APIs	Baserow Tables	Rankings, traffic, citations, leads flow back into measurement tables.
7. Optimization Loop	Baserow (threshold)	n8n Workflows	Performance below threshold triggers re-optimization workflows.

2. Baserow Data Layer: The Single Source of Truth

Baserow is the operational database backbone of the VirtualLab Digital ecosystem. Every keyword, every competitor, every content piece, every client interaction, and every performance metric lives in Baserow. The database is organized into 12 interconnected tables grouped into four functional domains. Each table has defined fields, field types, relationships to other tables, and standardized status fields that trigger n8n workflows. The following specification defines every table, every field, and every relationship with enough precision that a database administrator can implement the entire structure without ambiguity.

2.1 Domain 1: Keyword Intelligence (3 Tables)

2.1.1 Table: Keywords Master

This is the central keyword repository. Every keyword discovered through research, competitor analysis, or client input is stored here. Keywords are never deleted, only deprecated. Each record carries its complete research metadata including search intent classification, demand tier, competitive difficulty, and the specific page it maps to within the site architecture. The status field drives the content pipeline: when a keyword moves from "Research" to "Approved," it becomes eligible for content brief generation by the AI SEO Strategist agent.

Field Name	Field Type	Constraint	Description
Keyword ID	Auto Number	Unique	System-generated unique identifier.
Keyword	Text (Single Line)	Required	The exact search query string.
Search Volume (Monthly)	Number	Optional	Monthly estimated search volume from GSC or third-party tool.
CPC	Currency	Optional	Cost-per-click data for paid advertising reference.
Keyword Difficulty	Rating (1-100)	Optional	Competitive difficulty score (0=easiest, 100=hardest).
Search Intent	Single Select	Required	Informational / Commercial / Transactional / Navigational / Technical.
Buyer Awareness Stage	Single Select	Required	Unaware / Problem-Aware / Solution-Aware / Most-Aware.
Demand Tier	Single Select	Required	High / Medium / Emerging / Low / Seasonal.
Keyword Cluster	Link to Table	Required	Links to Keyword Clusters table.
Target Industry	Link to Table	Optional	Links to Industries table (HVAC, Roofing, etc.).
Target Page URL	Text (URL)	Required	The specific page this keyword maps to on the site.
Content Template	Single Select	Required	Hub Page / Service Detail / Resource Guide / Blog Post / FAQ / Industry Landing.
Current SERP Position	Number	Optional	Current organic ranking position (updated by monitoring workflow).
Current AI Citation Status	Single Select	Optional	Not Tracked / Not Cited / Cited (Positive) / Cited (Negative).
Status	Single Select	Required	Research / Clustered / Approved / In Production / Published / Deprecated.
Assigned To	Link to Table	Optional	Links to Team Members table.
Date Discovered	Date	Auto	When this keyword was added to the database.
Last Checked	Date	Auto	Last time SERP position was verified.
Notes	Long Text	Optional	Strategic notes, observations, or instructions.

2.1.2 Table: Keyword Clusters

Clusters group related keywords into topical entities. Each cluster represents a distinct theme that maps to a specific section of the website content architecture. Clusters prevent keyword cannibalization by establishing clear ownership boundaries. When the system detects that a new keyword might cannibalize an existing page, it

flags the record for human review before content production begins.

Field Name	Field Type	Constraint	Description
Cluster ID	Auto Number	Unique	System-generated unique identifier.
Cluster Name	Text	Required	e.g., "Core Agency Brand", "Google Maps and GBP", "GEO and AI Search".
Cluster Description	Long Text	Required	Detailed description of what this cluster covers and why it exists.
Primary Page URL	Text (URL)	Required	The hub page that owns this cluster.
Supporting Pages	Array of URLs	Optional	Spoke pages that support the hub.
Keywords Count	Formula (Count)	Auto	Count of linked keywords from Keywords Master table.
Avg Keyword Difficulty	Formula (Avg)	Auto	Average difficulty of all linked keywords.
Total Monthly Volume	Formula (Sum)	Auto	Sum of search volumes across all linked keywords.
Cannibalization Rule	Long Text	Required	Keywords to EXCLUDE from this cluster to prevent overlap.
Content Status	Single Select	Auto	No Content / In Progress / Published / Needs Update.
Last Updated	Date	Auto	Last time any keyword in this cluster was modified.

2.1.3 Table: SERP and LLM Citation Tracking

This table stores the results of systematic SERP monitoring and AI citation tracking. Every week, the n8n SERP Monitoring workflow queries the top 50 target keywords across Google, ChatGPT, Perplexity, and Google AI Overviews. The results are stored here with structured fields for position, citation context, and citation source. This data feeds the monthly Citation Dominance Score calculation and triggers optimization workflows when performance drops below defined thresholds.

Field Name	Field Type	Constraint	Description
Record ID	Auto Number	Unique	System-generated unique identifier.
Keyword	Link to Keywords Master	Required	Which keyword is being tracked.
Platform	Single Select	Required	Google Organic / Google Maps / ChatGPT / Perplexity / Gemini / AI Overviews.
VirtuaLab Mentioned	Checkbox	Required	Whether VirtuaLab Digital was mentioned in results.
Mention Context	Single Select	Required	Not Mentioned / Positive Citation / Neutral Mention / Negative / Competitor Only.
Citation Source URL	Text (URL)	Optional	The URL of the source that AI cited about VirtuaLab.
SERP Position	Number	Optional	Organic ranking position (1-100, or 0 if not found).
Featured Snippet	Checkbox	Optional	Whether a featured snippet appeared for this query.
AI Overview Present	Checkbox	Optional	Whether Google AI Overviews appeared.
Top Competitor Cited	Link to Competitors	Optional	Which competitor was cited instead of VirtuaLab.
Check Date	Date	Required	Date this check was performed.
Triggered Action	Single Select	Auto	None / Content Gap Alert / Optimization Workflow / Parasite Content Needed.

2.2 Domain 2: Content Operations (3 Tables)

2.2.1 Table: Content Calendar

Every piece of content, whether for the primary domain or a parasite platform, is tracked in the Content Calendar. This table is the operational backbone of the content pipeline. When a keyword moves to "Approved" status in Keywords Master, an n8n workflow automatically creates a draft Content Calendar record with a pre-filled content brief generated by the AI SEO Strategist agent. The content brief includes the target keyword, search intent, buyer awareness stage, recommended word count, internal linking targets, and GEO optimization requirements. Human editors review and approve briefs before content production begins.

Field Name	Field Type	Constraint	Description
Content ID	Auto Number	Unique	System-generated unique identifier.
Title	Text	Required	Working title of the content piece.
Content Type	Single Select	Required	Blog Post / Service Page / Industry Page / Resource Guide / FAQ / Parasite Article / YouTube Script / Social Post / GBP Post.
Target Platform	Single Select	Required	Primary Domain / LinkedIn / Medium / YouTube / Reddit / Quora / GBP.
Target Keyword	Link to Keywords Master	Required	Primary keyword this content targets.
Secondary Keywords	Array (Link)	Optional	Additional keywords to naturally incorporate.
Keyword Cluster	Link to Keyword Clusters	Required	Which cluster this content belongs to.
Target Industry	Link to Industries	Optional	Industry vertical if applicable.
Content Brief	Long Text	Required	Auto-generated by AI SEO Strategist. Includes outline, word count, GEO factors, internal links.
GEO Requirements	Long Text	Optional	Specific GEO factors to implement: statistics, citations, BLUF, entities.
Assigned Writer	Link to Team Members	Optional	Who is writing this content.
Assigned Client	Link to Clients	Optional	If this is client-specific content.
Word Count Target	Number	Required	Target word count for quality control.
Status	Single Select	Required	Brief Generated / Brief Approved / In Writing / In Review / Approved / Published / Archived.
Publish Date	Date	Required	Scheduled publication date.
Actual Publish Date	Date	Optional	Actual date content went live.
URL (Published)	Text (URL)	Optional	URL where content was published.
Internal Links Added	Checkbox	Auto	Whether internal links were verified after publication.
Schema Markup Added	Checkbox	Auto	Whether schema was added after publication.
Performance Check Date	Date	Auto	Date when 30-day performance review is scheduled.

2.2.2 Table: Competitor Intelligence

Every competitor identified in the strategy is tracked systematically. The Competitor Analysis Sentinel agent is responsible for maintaining this table with current data on competitor keywords, backlink profiles, content gaps, and pricing structures. When a competitor publishes new content or changes their pricing, the monitoring workflow updates this table and triggers an alert. This ensures that Virtualab always has current competitive intelligence without manual research efforts.

Field Name	Field Type	Constraint	Description
Competitor ID	Auto Number	Unique	System identifier.

Field Name	Field Type	Constraint	Description
Company Name	Text	Required	Competitor name.
Website URL	Text (URL)	Required	Primary domain.
Competitor Type	Single Select	Required	Direct Agency / Niche Specialist / Platform / Freelancer Network.
Target Verticals	Array (Link)	Optional	Which industries they target.
Pricing Model	Long Text	Optional	Known pricing tiers, minimums, contract terms.
Key Strengths	Long Text	Required	Documented strengths from analysis.
Key Weaknesses	Long Text	Required	Documented weaknesses and gaps.
Content Gap Opportunities	Long Text	Required	Specific content/topics they lack that VirtuaLab can exploit.
VirtuaLab Positioning	Long Text	Required	How VirtuaLab differentiates against this competitor.
Est. Domain Authority	Number	Optional	Estimated or measured domain authority.
Organic Keywords Tracked	Array (Link)	Optional	Keywords where this competitor ranks.
LLM Citation Frequency	Number	Optional	How often they appear in AI-generated responses.
Last Analyzed	Date	Auto	Last date of systematic competitive analysis.
Alert Active	Checkbox	Auto	Whether monitoring alerts are active for this competitor.

2.3 Domain 3: Client Operations (3 Tables)

2.3.1 Table: Clients

Field Name	Field Type	Constraint	Description
Client ID	Auto Number	Unique	System identifier.
Company Name	Text	Required	Client business name.
Industry Vertical	Link to Industries	Required	Primary industry.
Website URL	Text (URL)	Required	Client website domain.
GBP URL	Text (URL)	Optional	Google Business Profile URL.
Archetype	Single Select	Required	The Operator / The Practitioner / The Institution.
Annual Revenue Range	Single Select	Required	Under \$500K / \$500K-\$1M / \$1M-\$5M / \$5M-\$10M / \$10M-\$15M / Over \$15M.
Employee Count	Number	Optional	Approximate number of employees.
Monthly Ad Spend	Currency	Optional	Current monthly paid advertising budget.
Primary Pain Point	Single Select	Required	Paid Ad Fatigue / Maps Invisibility / Page 2 Syndrome / Leaky Leads / No SEO.
Service Package	Link to Service Packages	Required	Which VirtuaLab service package they purchased.
Contract Start Date	Date	Required	When the engagement began.
Contract End Date	Date	Required	When the engagement ends.
GHL Contact ID	Text	Optional	Linked GoHighLevel contact record.
Onboarding Status	Single Select	Required	Lead / Qualified / Proposal Sent / Contract Signed / Onboarding / Active / Churned.
Baseline Organic Traffic	Number	Optional	Monthly organic sessions at engagement start.
Baseline Map Pack Rate	Number	Optional	Percentage of target keywords in Map Pack at start.
Current Organic Traffic	Number	Auto	Latest monthly organic sessions (from GSC integration).
Current Map Pack Rate	Number	Auto	Latest Map Pack appearance percentage.
Lead Cost Baseline	Currency	Optional	Cost per lead at engagement start.
Current Lead Cost	Currency	Auto	Latest calculated cost per lead.

2.3.2 Table: Service Packages

Field Name	Field Type	Constraint	Description
Package ID	Auto Number	Unique	System identifier.
Package Name	Text	Required	e.g., "Local Visibility Starter", "Full AEO-Ready Website", "Multi-Location Dominance".
Description	Long Text	Required	What the package includes.
Price Range (Min)	Currency	Required	Minimum monthly or one-time fee.
Price Range (Max)	Currency	Required	Maximum monthly or one-time fee.
Billing Model	Single Select	Required	Monthly Retainer / One-Time Project / Hybrid.
Included Services	Array (Link)	Required	Links to Services table.
Deliverables	Long Text	Required	Specific deliverables per month/phase.
Contract Duration	Single Select	Required	Month-to-Month / 3 Months / 6 Months / 12 Months.
Active	Checkbox	Required	Whether this package is currently offered.

2.3.3 Table: Industries

Field Name	Field Type	Constraint	Description
Industry ID	Auto Number	Unique	System identifier.
Industry Name	Text	Required	e.g., HVAC, Plumbing, Roofing, Pest Control, Electrical, Dentistry, Landscaping.
Slug	Text	Required	URL slug: /industries/hvac.
Target Keywords Count	Formula	Auto	Count of keywords targeting this industry.
Active Clients Count	Formula	Auto	Count of active clients in this vertical.
Content Published Count	Formula	Auto	Count of published content pieces.
Average Client Revenue	Formula (Avg)	Auto	Average revenue of active clients in this vertical.
Seasonal Patterns	Long Text	Optional	Known seasonal demand patterns (e.g., HVAC peaks in summer).
Specific Search Challenges	Long Text	Optional	Vertical-specific search behavior notes.
Priority Level	Single Select	Required	Primary (active focus) / Secondary (established) / Emerging (new vertical).

2.4 Domain 4: Performance and Reporting (3 Tables)

2.4.1 Table: KPI Dashboard

Field Name	Field Type	Constraint	Description
KPI ID	Auto Number	Unique	System identifier.
Client	Link to Clients	Required	Which client this measures.
Metric Name	Single Select	Required	Organic Traffic / Map Pack Rate / Citation Score / Cost Per Lead / Review Velocity / Keyword Rankings.
Baseline Value	Number	Required	Value at contract start.
Current Value	Number	Auto	Latest measured value.
Target Value	Number	Required	Goal value for contract period.
Progress Percentage	Formula	Auto	$(\text{Current} - \text{Baseline}) / (\text{Target} - \text{Baseline}) \times 100$.
Measurement Date	Date	Auto	When this value was last measured.
Data Source	Single Select	Required	Google Search Console / BrightLocal / Manual AI Query / GHL CRM / Ahrefs.
Status	Formula	Auto	On Track (green) / At Risk (yellow) / Behind (red) - based on progress vs timeline.

2.4.2 Table: Parasite Platform Assets

Field Name	Field Type	Constraint	Description
Asset ID	Auto Number	Unique	System identifier.
Platform	Single Select	Required	LinkedIn / Medium / YouTube / Reddit / Quora / Google Sites / Cloud Property.
Profile URL	Text (URL)	Required	URL of the profile or property.
Platform Authority (DA)	Number	Optional	Domain authority of the platform.
Content Published Count	Formula	Auto	Count of content pieces published on this platform.
Total Impressions	Number	Auto	Aggregate impressions across all content on this platform.
Total Clicks	Number	Auto	Aggregate clicks to primary domain from this platform.
Last Published Date	Date	Auto	Most recent content publication date.
Publishing Cadence	Single Select	Required	Weekly / Bi-Weekly / Monthly / As Needed.
Status	Single Select	Required	Active / Paused / Needs Setup.

2.4.3 Table: Team Members

Field Name	Field Type	Constraint	Description
Member ID	Auto Number	Unique	System identifier.
Name	Text	Required	Team member name.
Role	Single Select	Required	SEO Strategist / Content Writer / Technical SEO / Account Manager / Web Developer.
Email	Email	Required	Contact email.
GHL User ID	Text	Optional	Linked GoHighLevel user.
Active Clients	Array (Link)	Auto	Clients assigned to this team member.
Capacity Status	Single Select	Required	Available / At Capacity / Overloaded.

Baserow Implementation Note: All tables use Baserow Link to Table fields for relationships. Formula fields are calculated automatically. Single Select fields use predefined options (no free-text for status/intent fields). Auto fields are system-generated. The n8n integration uses Baserow API webhooks for real-time triggering on record

creation and status changes.

3. n8n Automation Layer: The Nervous System

n8n is the automation engine that connects every component of the ecosystem. Without n8n, Baserow is a static database, GHL is a standalone CRM, and AI agents are isolated tools. With n8n, the entire ecosystem becomes a living system where data flows automatically between components, triggers actions based on business rules, and enforces process consistency. Every workflow defined below has a specific trigger, a defined process, and a measurable output. No workflow runs without a data trigger. No action is taken without a record being created or updated in Baserow.

3.1 Workflow 1: Keyword Research and Clustering Pipeline

This workflow automates the discovery, classification, and clustering of new keywords. It runs on a weekly schedule and also triggers on-demand when a new client onboards. The workflow pulls search data from the web search API, classifies each keyword by intent and awareness stage using an LLM call, checks for cannibalization against existing clusters in Baserow, and creates new records in the Keywords Master table with all fields pre-filled. The AI SEO Strategist agent processes the clustering logic.

Step	Action	Description
Trigger	Weekly Schedule (Monday 6AM) OR Manual Webhook	Starts the pipeline on schedule or on demand.
Step 1	Fetch Seed Keywords from Baserow	Pulls all keywords with Status = "Research" from Keywords Master.
Step 2	Expand Keywords via Search API	For each seed keyword, query search API for related terms, PAA questions, and variations.
Step 3	Classify Intent (LLM Call)	Send each keyword to AI SEO Strategist agent for intent/stage classification.
Step 4	Check Cannibalization	Compare new keywords against existing Keyword Clusters to detect overlap.
Step 5	Assign to Cluster	Based on intent + industry + cannibalization check, assign to appropriate cluster.
Step 6	Calculate Difficulty	Pull SERP data for top 10 results, estimate difficulty score.
Step 7	Write to Baserow	Create new records in Keywords Master with all fields populated.
Step 8	Update Cluster Counts	Recalculate keyword counts and avg difficulty for affected clusters.
Step 9	Alert if High-Value	If any keyword has High demand + Low difficulty + No content, send Slack/email alert.

3.2 Workflow 2: Content Brief Generation Pipeline

When a keyword moves to "Approved" status in Baserow, this workflow automatically generates a comprehensive content brief. The brief is created by the AI SEO Strategist agent and includes the target keyword, secondary keywords, recommended word count, H2/H3 outline, GEO optimization requirements, internal linking targets, and the specific buyer awareness stage messaging angle. The brief is written to the Content Calendar table as a new record with Status = "Brief Generated." A notification is sent to the assigned writer for review and approval before content production begins.

Step	Action	Description
Trigger	Baserow Webhook: Keywords Master.Status changes to Approved	Reactive trigger on keyword approval.
Step 1	Read Keyword Record	Fetch full keyword record including cluster, industry, intent, stage.
Step 2	Fetch Cluster Context	Pull all keywords in the same cluster to understand topical boundaries.
Step 3	Check Existing Content	Search Content Calendar for any existing content targeting this keyword.

Step	Action	Description
Step 4	Generate Brief (LLM Call)	AI SEO Strategist agent generates full content brief with outline.
Step 5	Add GEO Requirements	Based on content type, add specific GEO factors to implement.
Step 6	Create Content Calendar Record	Write brief to Content Calendar with Status = "Brief Generated".
Step 7	Notify Writer	Send notification via GHl task or email to assigned writer.

3.3 Workflow 3: SERP and AI Citation Monitoring

This workflow runs weekly and systematically checks the top 50 target keywords across six platforms: Google Organic, Google Maps, ChatGPT, Perplexity, Google Gemini, and Google AI Overviews. For each keyword-platform combination, the workflow records whether VirtuaLab was mentioned, the context of the mention, the ranking position, and which competitor was cited if VirtuaLab was not. This data flows into the SERP and LLM Citation Tracking table and feeds the monthly Citation Dominance Score report. When a tracked keyword drops more than 5 positions or loses an AI citation, the workflow triggers an optimization alert.

Step	Action	Description
Trigger	Weekly Schedule (Wednesday 6AM)	Systematic weekly check across all platforms.
Step 1	Fetch Target Keywords	Pull all keywords with Status = "Published" or "Approved" from Baserow.
Step 2	Query Google SERP	For each keyword, scrape top 10 organic results, record position.
Step 3	Query Google Maps	Check Map Pack for local intent keywords, record appearance.
Step 4	Query AI Platforms	Send target queries to ChatGPT, Perplexity, Gemini via API.
Step 5	Parse AI Responses	Extract mentions, context, and cited sources from AI responses.
Step 6	Write to Tracking Table	Create records in SERP and LLM Citation Tracking table.
Step 7	Calculate Citation Score	Compute monthly Citation Dominance Score percentage.
Step 8	Threshold Alert	If position drop > 5 or citation lost, create alert record.
Step 9	Trigger Optimization	For alerts, create optimization task in Content Calendar.

3.4 Workflow 4: Competitor Monitoring Alert

This workflow monitors the top 7 identified competitors (Hook Agency, Rival Digital, Blue Corona, Scorpion, RYNO Strategic, PlumberSEO.net, Sequoia GEO) for new content publications, pricing changes, and keyword movements. When a competitor publishes new content that targets keywords in VirtuaLab clusters, the workflow creates an alert in Baserow and optionally generates a response content brief. This ensures VirtuaLab can react quickly to competitive threats rather than discovering them months later during manual reviews.

Step	Action	Description
Trigger	Weekly Schedule (Friday 6AM)	Weekly competitor scan.
Step 1	Fetch Active Competitors	Pull all competitors with Alert Active = true from Competitor Intelligence.
Step 2	Scrape Competitor Blogs	Check for new blog posts or content pages published since last check.
Step 3	Extract Target Keywords	Parse competitor content for target keyword usage.
Step 4	Check for Overlap	Cross-reference competitor keywords with VirtuaLab keyword clusters.
Step 5	Monitor Pricing Pages	Scrape pricing pages for changes in tiers or minimums.
Step 6	Generate Alert	If overlap or pricing change detected, create alert in Baserow.
Step 7	Optional: Generate Response Brief	If high-priority overlap, auto-generate response content brief.
Step 8	Update Competitor Record	Update Last Analyzed date and citation frequency.

3.5 Workflow 5: Client Reporting Pipeline

This workflow automates the generation of monthly client performance reports. On the first business day of each month, the workflow pulls KPI data from Google Search Console, BrightLocal (for local rankings), the SERP/LLM Citation Tracking table, and the GHL CRM (for lead data). It calculates month-over-month changes, computes progress toward targets, and generates a structured report that is uploaded to GHL as a PDF and sent to the client via email. The report includes organic traffic trends, Map Pack appearance rates, keyword ranking changes, AI citation progress, and cost-per-lead calculations. This eliminates the manual report-writing process and ensures every client receives a consistent, data-rich monthly report.

Step	Action	Description
Trigger	Monthly Schedule (1st of month, 7AM)	Automated monthly reporting.
Step 1	Fetch Active Clients	Pull all clients with Onboarding Status = "Active" from Baserow.
Step 2	Pull GSC Data	For each client, query Google Search Console API for last 30 days.
Step 3	Pull Local Data	Query BrightLocal API for Map Pack and local ranking data.
Step 4	Pull Citation Data	Aggregate SERP/LLM Citation Tracking data for the month.
Step 5	Pull Lead Data	Query GHL CRM for leads generated, conversion rate, cost per lead.
Step 6	Calculate KPIs	Compute all KPI values and update KPI Dashboard table in Baserow.
Step 7	Generate Report (LLM)	AI agent generates narrative analysis of the data.
Step 8	Create PDF Report	Render report as PDF using template system.
Step 9	Upload to GHL	Attach PDF to client contact record in GoHighLevel.
Step 10	Send Email	Send report email to client with PDF attachment.

3.6 Workflow 6: Content Publishing and Verification

When content moves to "Approved" status in the Content Calendar, this workflow manages the publication process. For primary domain content, it triggers a notification for the web developer to publish. For parasite platform content, it generates platform-specific formatting and posts via API where possible (LinkedIn, Medium). After publication, the workflow verifies that the content is live, checks that internal links are functional, confirms schema markup is present, and updates the content record with the published URL and date. It also schedules a 30-day performance review by creating a future-dated task in the system.

3.7 n8n Workflow Summary

Workflow	Frequency	Trigger	Writes To	Purpose
WF-01: Keyword Research and Clustering	Weekly + On-Demand	Baserow Webhook	Keywords Master	Auto-discover, classify, and cluster keywords.
WF-02: Content Brief Generation	Reactive (on keyword approval)	Baserow Webhook	Content Calendar	Generate AI content briefs for approved keywords.
WF-03: SERP and AI Citation Monitoring	Weekly	Schedule	SERP/LLM Tracking	Track rankings and AI citations across 6 platforms.
WF-04: Competitor Monitoring	Weekly	Schedule	Competitor Intelligence	Monitor competitors for new content and pricing changes.
WF-05: Client Reporting	Monthly	Schedule	KPI Dashboard + GHL	Generate and deliver monthly performance reports.
WF-06: Content Publishing	Reactive (on content approval)	Baserow Webhook	Content Calendar	Manage publication and post-publish verification.

Workflow	Frequency	Trigger	Writes To	Purpose
WF-07: Review Generation Nudge	Bi-Weekly	Schedule	GHL Tasks	Nudge clients to request Google reviews from customers.
WF-08: GBP Post Scheduling	Bi-Weekly	Schedule	Content Calendar	Schedule and track Google Business Profile posts.

4. GoHighLevel CRM Layer: Client Delivery Pipeline

GoHighLevel (GHL) serves as the client-facing operational layer of the ecosystem. While Baserow is the internal data backbone and n8n is the automation engine, GHL is where client interactions happen: lead capture, proposal delivery, contract signing, onboarding checklists, task management, communication, and reporting. The GHL pipeline is designed to mirror the buyer journey defined in the strategy document, with pipeline stages that correspond to each step from initial awareness through active delivery and renewal.

4.1 Pipeline Stages

Pipeline Stage	Definition	Automated Actions
Stage 1: New Lead	Contact created via website form, diagnostic audit request, or referral. No qualification yet.	Auto-respond with diagnostic audit delivery (if applicable). Add to nurture sequence.
Stage 2: Qualified	Lead has engaged with content, requested audit, or met qualification criteria (\$1M+ revenue, local service, paid ad dependency).	Trigger n8n to create Client record in Baserow. Assign account manager.
Stage 3: Proposal Sent	Custom proposal generated based on industry vertical, pain points, and service needs. Sent via GHL email/document system.	Proposal includes specific deliverables, timeline, pricing, and case study references. Track open rate.
Stage 4: Contract Signed	Client has signed the engagement contract and paid initial invoice.	Trigger n8n onboarding workflow: create Baserow records, set up GSC access, schedule kick-off call.
Stage 5: Onboarding (Week 1-2)	Technical audit completed. Access established. Baseline metrics recorded in Baserow. KPI Dashboard initialized.	Complete onboarding checklist: GSC, GBP access, CRM integration, baseline report.
Stage 6: Active Delivery	Ongoing SEO, content, and optimization work. Monthly reports delivered. Weekly tasks assigned and tracked.	n8n workflows drive content pipeline, reporting, and monitoring automatically.
Stage 7: Renewal (60 days before end)	Renewal conversation initiated. Performance data compiled. ROI presentation prepared.	Generate ROI summary from Baserow KPI Dashboard. Prepare renewal proposal.
Stage 8: Churned / Lost	Client did not renew or terminated early. Reason recorded for analysis.	Log churn reason in Baserow. Trigger post-mortem analysis workflow.

4.2 GHL Automation Rules

GHL workflows complement n8n automations by handling the client communication and task management layer. When a new lead enters Stage 1, GHL automatically sends a welcome email with the diagnostic audit results (if applicable) and adds the contact to a 5-email nurture sequence. The nurture sequence is paced over 14 days and includes educational content about local SEO, paid ad dependency, and GEO. Each email includes a tracking pixel that updates the lead score in GHL based on opens and clicks. When a lead score reaches a defined threshold, GHL automatically moves the contact to Stage 2 (Qualified) and assigns an account manager through a round-robin distribution rule.

During Active Delivery (Stage 6), GHL task management ensures that every weekly deliverable is tracked. n8n creates GHL tasks for content publication, review requests, and reporting deadlines. Each task has a due date, an assignee, and a checklist of completion criteria. When a task is marked complete, GHL triggers a webhook to n8n, which updates the corresponding Baserow record and proceeds to the next step in the workflow. This bidirectional integration between GHL and Baserow (mediated by n8n) ensures that client-facing task management and internal data management stay perfectly synchronized.

4.3 GHL Dashboard Configuration

The GHL dashboard is configured to show account managers a real-time view of their client portfolio. The primary dashboard displays each active client with key metrics: organic traffic trend (sparkline), Map Pack appearance rate, citation score, leads this month, and cost per lead. Color coding (green/yellow/red) reflects KPI status from the Baserow KPI Dashboard table. A secondary dashboard shows the content pipeline: how many briefs are pending, in writing, in review, and published this month. A third dashboard shows parasite platform performance: impressions, clicks, and citation contributions from each external platform. All dashboard data is pulled from Baserow via n8n API calls, ensuring a single source of truth.

5. AI Agent Orchestration Layer

The AISEO Framework defines six specialized AI agents that operate within the ecosystem. Each agent has a distinct role, a defined input/output specification, and a clear integration point with the Baserow/n8n infrastructure. Agents are not autonomous decision-makers; they are specialized processors that receive structured inputs from Baserow (via n8n), apply their expertise, and return structured outputs that are written back to Baserow. Human operators review agent outputs before they trigger client-facing actions. This ensures quality control while leveraging AI speed and consistency for data-heavy tasks.

5.1 Agent Specifications

Agent	Responsibilities	Input Sources (Baserow)	Output Destinations	n8n Integration	Operating Rules
1. Deep Marketing Strategist	Market analysis, buyer psychology, demand modeling, competitive positioning	Baserow: Clients table, Industries table, Competitor Intelligence table	Baserow: Updates client ICP notes, industry demand patterns, positioning recommendations	n8n WF-01 (keyword expansion), WF-04 (competitor analysis)	LLM system prompt includes full ICP data and psychographic profiles from strategy doc. Outputs must include data citations.
2. Positioning Strategist	Brand messaging, differentiation analysis, value proposition development, content angle creation	Baserow: Competitor Intelligence, Keywords Master (commercial intent), Content Calendar	Baserow: Content briefs with messaging angles, competitor differentiation notes	n8n WF-02 (content brief generation)	Inputs include competitor strengths/weaknesses. Outputs must specify unique angles vs each competitor.
3. Competitor Analysis Sentinel	SERP analysis, competitor content monitoring, backlink profiling, gap identification	Baserow: Competitor Intelligence, SERP/LLM Citation Tracking, Keywords Master	Baserow: Competitor records, gap opportunities, alert triggers	n8n WF-03 (SERP monitoring), WF-04 (competitor alerts)	Must track all 7 primary competitors. Output must include specific URL-level gaps.
4. AI SEO Strategist	Keyword clustering, content brief generation, GEO optimization, topic mapping	Baserow: Keywords Master, Keyword Clusters, Content Calendar	Baserow: Keyword clusters, content briefs, GEO requirements	n8n WF-01 (clustering), WF-02 (briefs)	Core agent for content pipeline. Must implement all 9 GEO factors in briefs.
5. Parasite SEO Strategist	Platform selection, content formatting per platform, publishing cadence, citation tracking	Baserow: Parasite Platform Assets, Content Calendar (parasite type), SERP/LLM Tracking	Baserow: Platform-specific content guidelines, publication calendar, performance tracking	n8n WF-06 (publishing), WF-03 (citation tracking)	Must know platform-specific DA, audience, and content format rules.
6. Sentinel (Quality Gate)	Output validation, quality scoring, cannibalization checking, brand voice compliance	Baserow: All tables (read-only audit)	Baserow: Quality scores, validation flags, required revision notes	All workflows (post-processing step)	Final gate before any content reaches human review or publication. Rejects content that violates brand rules.

5.2 Agent Communication Protocol

Agents communicate through Baserow, not directly with each other. When the AI SEO Strategist generates a content brief, it writes the brief to the Content Calendar table. The Sentinel agent then reads the brief from Baserow, validates it against quality rules, and writes a quality score and any revision notes back to the same record. This indirect communication pattern ensures that every agent interaction is logged, auditable, and recoverable. If an agent produces an error, the record in Baserow shows exactly what went wrong and which agent was responsible.

The n8n workflow acts as the orchestrator that sequences agent calls. For example, in the content brief generation workflow (WF-02), n8n first calls the AI SEO Strategist to generate the brief, then calls the Sentinel to validate it. If the Sentinel approves, the record status moves to "Brief Generated" and a notification is sent. If the Sentinel rejects, the record stays in draft and a revision task is created. This sequential, quality-gated approach ensures that no substandard output reaches human reviewers or clients.

5.3 Agent Input/Output Schema

Every agent interaction follows a standardized JSON schema that is stored in Baserow Long Text fields. This ensures consistency and enables the Sentinel agent to validate outputs programmatically. The schema includes: `agent_id` (which agent produced this), `input_data` (the structured inputs received), `output_data` (the structured response), `quality_score` (0-100, assigned by Sentinel), `validation_flags` (array of issues found), `timestamp`, and `processing_time_ms`. This structured logging enables performance tracking across agents, identification of which agents produce the highest-quality outputs, and continuous improvement of agent prompts based on quality data.

6. Measurement and Feedback Protocol

The measurement protocol defines exactly what is measured, how often, by which tool, and what actions are triggered by the results. This is the layer that closes the feedback loop and transforms the ecosystem from a static system into a self-improving machine. Every metric has a defined data source, collection frequency, storage location (Baserow table), threshold for action, and automated response workflow. There is no metric that is tracked without a corresponding action trigger. If a metric is worth measuring, it is worth acting on when it crosses a threshold.

6.1 Measurement Cadence

Metric	Data Source	Frequency	Stored In	Action Threshold	Triggered Response
Organic Traffic	Google Search Console API	Daily (auto-pull)	KPI Dashboard	30-day trend drops > 15%	n8n WF-05 + Content Calendar alert
Keyword Rankings (Top 50)	Ahrefs / SEMrush API	Bi-Weekly	Keywords Master.SERP Position	Position drop > 5 positions	n8n WF-03 + optimization task
Map Pack Appearance	BrightLocal API	Weekly	KPI Dashboard	Drop below 30% for target suburbs	n8n WF-05 + GBP optimization task
AI Citation Score	Manual AI Query (via n8n LLM calls)	Monthly	SERP/LLM Tracking	Score drops below 15%	n8n WF-03 + parasite content task
Cost Per Lead	GHL CRM (leads / ad spend)	Monthly	KPI Dashboard	Increase > 20% from baseline	Account manager alert in GHL
Review Velocity	Google Business Profile API	Bi-Weekly	KPI Dashboard	Below 3 reviews/month	n8n WF-07 (review nudge workflow)
Competitor Keywords	Ahrefs API	Monthly	Competitor Intelligence	New competitor keyword in our cluster	n8n WF-04 + content response brief
Content Performance	GSC + Baserow	Monthly (30-day post-publish)	Content Calendar	0 organic impressions after 30 days	Content revision task
Parasite Impressions	Platform Analytics APIs	Monthly	Parasite Platform Assets	Below 1K monthly impressions	Platform content refresh task
Client Satisfaction	GHL (manual or survey)	Quarterly	Clients table (notes field)	NPS score below 7	Account manager escalation

6.2 Weekly Review Protocol

Every Monday, the system generates an automated Weekly Operations Digest that is sent to all team members via GHL. The digest includes: total keywords in pipeline (by status), content published this week, content due next week, SERP position changes for tracked keywords, new competitor content detected, AI citation mentions gained/lost, and any alerts requiring human attention. This digest is generated by n8n pulling data from Baserow and formatting it through an LLM call that adds narrative context. The review ensures that every team member starts the week with complete operational awareness without manually checking multiple systems.

6.3 Monthly Strategic Review

On the first Wednesday of every month, a strategic review is conducted that goes beyond operational metrics. This review examines: keyword cluster performance trends (which clusters are gaining/losing authority), content

ROI analysis (which content types generate the most business impact), competitor landscape shifts (new entrants, positioning changes), AI citation trajectory (month-over-month growth rate), and client health scores (aggregated NPS and retention indicators). The review is documented in Baserow as a Strategic Review record and drives adjustments to the keyword strategy, content calendar, and resource allocation for the following month. This is the mechanism by which the system adapts and improves over time based on measured outcomes rather than assumptions.

7. Implementation Sequence

The ecosystem must be built in a specific sequence to ensure that each layer functions correctly before the next layer depends on it. Building the n8n workflows before Baserow tables exist would fail. Building the GHL pipeline before the Baserow data model is defined would create orphaned client records. The implementation sequence below ensures that each step creates the prerequisite for the next, minimizing rework and ensuring that the system is functional at every stage.

Phase	Timeline	Actions	Success Criteria
Phase 1: Foundation	Week 1-2	Create all 12 Baserow tables with exact field specifications from Chapter 2. Configure all Link to Table relationships. Set up Single Select options for all status/intent/type fields. Create formula fields for auto-calculations. Test with 10 sample keyword records.	Baserow is operational with complete schema.
Phase 2: Core Workflows	Week 3-4	Build n8n WF-01 (Keyword Research) and WF-02 (Content Brief Generation) end-to-end. Test with 5 real keywords from the strategy. Verify Baserow records are created correctly. Verify LLM agent calls return valid outputs. Verify Sentinel quality gate works.	Keywords can be researched and content briefs generated automatically.
Phase 3: Monitoring	Week 5-6	Build n8n WF-03 (SERP/AI Monitoring) and WF-04 (Competitor Monitoring). Configure weekly schedules. Run first full monitoring cycle across 50 keywords and 7 competitors. Verify tracking data flows into Baserow correctly.	System can track rankings, citations, and competitors automatically.
Phase 4: Client Delivery	Week 7-8	Configure GHL pipeline stages (8 stages). Build n8n WF-05 (Client Reporting) and WF-06 (Content Publishing). Set up GHL automation rules (nurture sequences, task creation, round-robin assignment). Test with 1 internal client (VirtualLab itself).	Full client lifecycle from lead to reporting is automated.
Phase 5: Integration	Week 9-10	Build remaining workflows (WF-07 Review Nudge, WF-08 GBP Posts). Connect all bidirectional integrations (Baserow <-> n8n <-> GHL). Run full system test with 5 simulated client scenarios. Fix integration issues.	All 8 workflows operational. All 4 layers communicating.
Phase 6: Data Population	Week 11-12	Import all keywords from the strategy document into Baserow. Enter all 7 competitors. Create parasite platform asset records. Set up team member records. Initialize KPI baselines for first client.	System is fully populated with real data and ready for production use.
Phase 7: Go-Live	Week 13+	Begin operating the system for real client delivery. Monitor for 4 weeks. Collect feedback from team. Iterate on workflows, agent prompts, and dashboard configurations based on actual usage patterns.	System is production-ready and self-improving.

7.1 System Dependencies Map

Component	Depends On	Rationale
Baserow Tables	None	Everything else depends on data structures being defined first.
n8n Workflows	Baserow Tables	Workflows read from and write to Baserow. Cannot function without tables.
AI Agent Prompts	Baserow schema + Strategy Doc	Agent prompts reference exact field names and table structures.
GHL Pipeline	Baserow Clients table	GHL pipeline creates records that sync back to Baserow.

Component	Depends On	Rationale
GHL Automations	GHL Pipeline stages	Automation rules trigger based on pipeline stage changes.
n8n-GHL Integration	Both n8n and GHL configured	Bidirectional sync requires both systems operational.
Client Reporting	All above	Reports pull from Baserow, format via n8n, deliver via GHL.
Measurement Protocol	Baserow KPI Dashboard + n8n	Threshold alerts require Baserow data and n8n trigger logic.